Supplementary File for Paper: Confidence-Guided Deep Reinforcement Learning Algorithm for Stochastic Energy-Minimization Scheduling in Flexible 3C Job Shop

Supplementary Proof

Proposition: Under the adopted Gamma-based approximation of candidate-solution objective uncertainty, the improved optimal computing budget allocation (ICBA) rule satisfies:

$$\frac{I_u^t}{I_h^t} = \frac{(\nu_u^a)^2[(\nu_b^a)^2\hat{\alpha}_u - (\nu_u^a)^2\hat{\alpha}_b]^2}{(\nu_h^a)^2[(\nu_b^a)^2\hat{\alpha}_h - (\nu_h^a)^2\hat{\alpha}_b]^2} \tag{1}$$

$$I_b^t = \sum_{u=1,u\neq b}^{N_p} \frac{(\nu_b^a)^2}{(\nu_u^a)^2} I_u^t \tag{2}$$

where $\hat{\alpha}_u$, $\hat{\alpha}_h$, and $\hat{\alpha}_b$ are the shape parameters in the adopted Gamma-based approximation of the prior objective uncertainty of candidate solutions x_u, x_h, and x_b, respectively. Similarly, ν_u^a, ν_h^a, and ν_b^a denote the sample standard deviations computed from the deviations of the current simulation outputs from the corresponding prior mean objective estimates of candidate solutions x_u, x_h, and x_b, respectively.

Proof: Let b denote the index of the reference best candidate solution selected according to the prior mean objective estimate, i.e., $b = \arg\min_{1\leq u\leq N_p} \hat{J}_u$, which is consistent with the allocation setting in the main text. The OCBA allocation problem aims to maximize the approximate probability of correct selection. Equivalently, it can be formulated as follows:

$$1 - \sum_{u=1,u\neq b}^{N_p} \mathbb{P}\{\tilde{J}_b > \tilde{J}_u\} \tag{3}$$

$$s.t. \sum_{u=1}^{N_p} I_u = I, u = 1, \cdots, N_p \tag{4}$$

In this supplementary derivation, I denotes the total evaluation budget, corresponding to B^t in the main text. The variable $\tilde{J}_u$ in (3) denotes the updated objective model of candidate solution x_u, obtained by combining prior objective information with current simulation outputs. For tractable objective updating and budget allocation, the uncertainty in each candidate solution's objective estimate is approximated by a Gamma-family model. Under this approximation, the updated objective model is written as:

$$\text{Gamma}(\hat{J}_u \mid \hat{\alpha}_u, \beta) = \frac{\beta^{\hat{\alpha}_u}}{\Gamma(\hat{\alpha}_u)} (\hat{J}_u)^{\hat{\alpha}_u - 1} \exp(-\beta \hat{J}_u) \tag{5}$$

where $\hat{\alpha}_u$ is the shape parameter in the adopted Gamma-based approximation of the prior objective uncertainty of candidate solution x_u, calculated by:

$$\hat{\alpha}_u = \max_{k=1,\cdots,m} [\sum_{i=1}^{n} \frac{w^u_{(i,k)}}{\bar{p}_{(i,k)}\theta^2}] \tag{6}$$

where $w^u_{(i,k)} = 1$ if job i is processed on machine k under candidate solution x_u, and $w^u_{(i,k)} = 0$ otherwise. Here, $\hat{\alpha}_u$ is used as the shape parameter in the adopted Gamma approximation of the prior objective uncertainty of candidate solution x_u. In addition, $\exp(\cdot)$ denotes the natural exponential function, and $\Gamma(\cdot)$ denotes the Gamma function, i.e.,

$$\Gamma(\alpha) = \int_0^\infty v^{\alpha-1} \exp(-v) dv \tag{7}$$

According to [1], the updated objective model $\tilde{J}_u$ of candidate solution x_u under the adopted approximation can be written as:

$$\tilde{J}_u \sim \text{Gamma}\left(\hat{\alpha}_u + \frac{I_u}{2}, \beta + \frac{I_u(\nu^a_u)^2}{2}\right) \tag{8}$$

To simplify the notation, we rewrite $\hat{\alpha}_u + \frac{I_u}{2}$ and $\beta + \frac{I_u(\nu^a_u)^2}{2}$ as α_u and β_u, respectively. According to the definition of the objective function, we have:

$$\begin{aligned} &\sum_{u=1,u\neq b}^{N_p} \mathbb{P}(\tilde{J}_b > \tilde{J}_u) \\ =& \sum_{u=1,u\neq b}^{N_p} \int_{-\infty}^{\infty} \int_q^{\infty} \frac{\beta_b^{\alpha_b}}{\Gamma(\alpha_b)} v^{\alpha_b-1} \exp(-\beta_b v) \\ &\cdot \frac{\beta_u^{\alpha_u}}{\Gamma(\alpha_u)} q^{\alpha_u-1} \exp(-\beta_u q) dv dq \end{aligned} \tag{9}$$

Let $v = yw$ and $q = y(\frac{1-\beta_b w}{\beta_u})$. The Jacobian determinant of the transformation in Eq. (9) is $-\frac{y}{\beta_u}$, and it can be rewritten as:

$$\begin{aligned} &\sum_{u=1,u\neq b}^{N_p} \mathbb{P}(\tilde{J}_b > \tilde{J}_u) \\ =& \sum_{u=1,u\neq b}^{N_p} \frac{\beta_b^{\alpha_b}\beta_u^{\alpha_u}}{\Gamma(\alpha_b)\Gamma(\alpha_u)} \int_0^\infty \int_{\frac{1}{\beta_b}}^{\frac{1}{\beta_b+\beta_u}} (yw)^{\alpha_b-1} [y(\frac{1-\beta_b w}{\beta_u})]^{\alpha_u-1} \\ &\cdot \exp(-y)(-\frac{y}{\beta_u}) dy dw \\ =& \sum_{u=1,u\neq b}^{N_p} \frac{\beta_b^{\alpha_b}\beta_u^{\alpha_u}}{\Gamma(\alpha_b)\Gamma(\alpha_u)} \int_0^\infty y^{\alpha_b+\alpha_u-1} \exp(-y) dy \\ &\cdot \int_{\frac{1}{\beta_b+\beta_u}}^{\frac{1}{\beta_b}} \frac{w^{\alpha_b-1}(1-\beta_b w)^{\alpha_u-1}}{\beta_u^{\alpha_u}} dw \end{aligned}$$

$$
\begin{aligned}
&= \sum_{u=1,u\neq b}^{N_p} \frac{\beta_b}{B(\alpha_u,\alpha_b)} \int_{\frac{1}{\beta_b+\beta_u}}^{\frac{1}{\beta_b}} (\beta_b w)^{\alpha_b-1}(1-\beta_b w)^{\alpha_u-1} dw \\
&= \sum_{u=1,u\neq b}^{N_p} \frac{1}{B(\alpha_u,\alpha_b)} \int_{\frac{\beta_b}{\beta_b+\beta_u}}^{1} r^{\alpha_b-1}(1-r)^{\alpha_u-1} dr \\
&= \sum_{u=1,u\neq b}^{N_p} \frac{1}{B(\alpha_u,\alpha_b)} \int_{0}^{\frac{\beta_u}{\beta_b+\beta_u}} t^{\alpha_u-1}(1-t)^{\alpha_b-1} dt \\
&= \sum_{u=1,u\neq b}^{N_p} \frac{B_I(\alpha_u,\alpha_b)}{B(\alpha_u,\alpha_b)}
\end{aligned} \tag{10}
$$

where $B(\cdot,\cdot)$ denotes the Euler beta function, and $B_I(\cdot,\cdot)$ denotes the corresponding incomplete beta function, defined as:

$$
\begin{aligned}
B(\alpha_u,\alpha_b) &= \int_0^1 t^{\alpha_u-1}(1-t)^{\alpha_b-1} dt \\
B_I(\alpha_u,\alpha_b) &= \int_0^{\frac{\beta_u}{\beta_b+\beta_u}} t^{\alpha_u-1}(1-t)^{\alpha_b-1} dt
\end{aligned} \tag{11}
$$

The above expression corresponds to the cumulative probability of a beta distribution. Let $b_u = \frac{\beta_u}{\beta_b+\beta_u}$. Since $\Delta = 4$, and I_u is an even number in this paper, α_b and α_u are positive integers. Thus, we can calculate Eq. (10) by repeatedly using integration by parts and obtain:

$$
\begin{aligned}
&\sum_{u=1,u\neq b}^{N_p} \mathbb{P}(\tilde{J}_b > \tilde{J}_u) \\
&= \sum_{u=1,u\neq b}^{N_p} \sum_{j=\alpha_u}^{\alpha_b+\alpha_u-1} \binom{\alpha_b+\alpha_u-1}{j} b_u^j b_u^{\alpha_b+\alpha_u-1-j}
\end{aligned} \tag{12}
$$

where $\binom{\alpha_b+\alpha_u-1}{j}$ denotes the binomial coefficient. This expression establishes the connection between the OCBA objective and a binomial tail probability. The Laplace approximation [2] provides a way to approximate this binomial tail probability by a normal distribution when $b_u(\alpha_b+\alpha_u-1)$ or $(1-b_u)(\alpha_b+\alpha_u-1)$ is large enough (larger than 5, as commonly required for the normal approximation). In this paper, $\hat{\alpha} = \sum 1/(\overline{p}_{(i,k)} \cdot \theta^2)$ is on the order of 10^5. Thus, both $b_u(\alpha_b+\alpha_u+1)$ and $(1-b_u)(\alpha_b+\alpha_u+1) \gg 5$, which enables us to follow from [2] and obtain:

$$
\begin{aligned}
\sum_{u=1,u\neq b}^{N_p} \mathbb{P}(\tilde{J}_b > \tilde{J}_u) &\approx \sum_{u=1,u\neq b}^{N_p} \int_{\alpha_u}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_u} \exp(-\frac{(v-\delta_u)^2}{2\sigma_u^2}) dv \\
&= \sum_{u=1,u\neq b}^{N_p} \int_{\frac{\alpha_u-\delta_u}{\sigma_u}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-\frac{t^2}{2}) dt
\end{aligned} \tag{13}
$$

where $\delta_u = ...$ and $\sigma_u = ...$ are introduced for notational simplicity. Let λ be the Lagrange multiplier

associated with the budget constraint. Then, we have:

$$F = 1 - \sum_{u=1,u\neq b}^{N_p} \int_{\frac{\alpha_u-\delta_u}{\sigma_u}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-\frac{t^2}{2})dt - \lambda(\sum_{u}^{N_p} I_u - I) \tag{14}$$

Hence, noting that $\alpha_b + \alpha_u \gg 1$, the KKT conditions [3] of (14) can be stated as:

$$\begin{aligned}
\frac{\partial F}{\partial I_u} &= \frac{\partial F}{\partial(\frac{\alpha_u-\delta_u}{\sigma_u})} \left[\frac{\partial(\frac{\alpha_u-\delta_u}{\sigma_u})}{\partial \alpha_u} \frac{\partial \alpha_u}{\partial I_u} + \frac{\partial(\frac{\alpha_u-\delta_u}{\sigma_u})}{\partial \beta_u} \frac{\partial \beta_u}{\partial I_u} \right] - \lambda \\
&\approx -\lambda - \frac{1}{2\sqrt{2\pi}} \exp\left[-\frac{(\frac{\alpha_u-\delta_u}{\sigma_u})^2}{2} \right] \\
&\quad \cdot \left[\frac{2\alpha_b\beta_b + \alpha_u\beta_b + \alpha_b\beta_u}{4(\alpha_u+\alpha_b)\sqrt{\beta_b\beta_u(\alpha_u+\alpha_b)}} - \frac{e_u^2(\alpha_b\beta_u + \alpha_u\beta_b)}{4\beta_u\sqrt{\beta_b\beta_u(\alpha_u+\alpha_b)}} \right] \\
&= 0, \ \forall u = 1, 2, \cdots, N_p, \text{ and } u \neq b,
\end{aligned} \tag{15}$$

$$\begin{aligned}
\frac{\partial F}{\partial I_b} &= \frac{\partial F}{\partial(\frac{\alpha_u-\delta_u}{\sigma_u})} \left[\frac{\partial(\frac{\alpha_u-\delta_u}{\sigma_u})}{\partial \alpha_b} \frac{\partial \alpha_b}{\partial I_b} + \frac{\partial(\frac{\alpha_u-\delta_u}{\sigma_u})}{\partial \beta_b} \frac{\partial \beta_b}{\partial I_b} \right] - \lambda \\
&\approx -\lambda - \frac{1}{2\sqrt{2\pi}} \sum_{u=1,u\neq b}^{N_p} \exp\left[-\frac{(\frac{\alpha_u-\delta_u}{\sigma_u})^2}{2} \right] \\
&\quad \cdot \left[-\frac{2\alpha_u\beta_u + \alpha_u\beta_b + \alpha_b\beta_u}{4(\alpha_u+\alpha_b)\sqrt{\beta_b\beta_u(\alpha_u+\alpha_b)}} + \frac{e_b^2(\alpha_u\beta_b + \alpha_b\beta_u)}{4\beta_b\sqrt{\beta_b\beta_u(\alpha_u+\alpha_b)}} \right] \\
&= 0,
\end{aligned} \tag{16}$$

and

$$\lambda(\sum_{u=1}^{N_p} I_u - I) = 0, \text{ and } \lambda \geq 0 \tag{17}$$

As discussed in [4], OCBA focuses on asymptotically optimal allocation. Therefore, we consider the asymptotic case where $I \to \infty$. From Eqs. (15)-(16), we have:

$$\sum_{u=1,u\neq b}^{N_p} \frac{\lambda(\nu_b^a)^2 I_u}{(\nu_u^a)^2 I_b} - \lambda = 0 \tag{18}$$

Then,

$$I_b = \sum_{u=1,u\neq b}^{N_p} \frac{(\nu_b^a)^2}{(\nu_u^a)^2} I_u \tag{19}$$

We next consider the relationship between I_u and I_h, $\forall u, q \in \{1, 2, \cdots, N_p\}$ and $u \neq q \neq b$ from Eq. (15). We have:

$$
\begin{aligned}
&\left[\frac{2\alpha_b\beta_b+\alpha_u\beta_b+\alpha_b\beta_u}{4(\alpha_u+\alpha_b)\sqrt{\beta_b\beta_u(\alpha_u+\alpha_b)}}-\frac{e_u^2(\alpha_b\beta_u+\alpha_u\beta_b)}{4\beta_u\sqrt{\beta_b\beta_u(\alpha_u+\alpha_b)}}\right]\\
&\times\exp\left[-\frac{(\frac{\alpha_u-\delta_u}{\sigma_u})^2}{2}\right]=\exp\left[-\frac{(\frac{\alpha_h-\delta_h}{\sigma_h})^2}{2}\right]\\
&\times\left[\frac{2\alpha_b\beta_b+\alpha_h\beta_b+\alpha_b\beta_h}{4(\alpha_u+\alpha_b)\sqrt{\beta_b\beta_h(\alpha_h+\alpha_b)}}-\frac{e_h^2(\alpha_b\beta_h+\alpha_h\beta_b)}{4\beta_h\sqrt{\beta_b\beta_h(\alpha_h+\alpha_b)}}\right]
\end{aligned}
\tag{20}
$$

According to Eq. (19), we can assume that $I_b \gg I_u$. By rearranging terms of Eq. (20), and taking the natural log on both sides, we have:

$$
\begin{aligned}
&\frac{\alpha_u\beta_b-\alpha_b\beta_u}{\sqrt{\beta_u}}+\ln\frac{2\alpha_b\beta_b\beta_u-e_u^2\alpha_b(\alpha_b\beta_u+\alpha_u\beta_b)}{\sqrt{\beta_u}}\\
=&\frac{\alpha_h\beta_b-\alpha_b\beta_h}{\sqrt{\beta_h}}+\ln\frac{2\alpha_b\beta_b\beta_h-e_h^2\alpha_b(\alpha_b\beta_h+\alpha_h\beta_b)}{\sqrt{\beta_u}}
\end{aligned}
\tag{21}
$$

Since all the log terms increase much slower than others, we can obtain the asymptotic relationship between I_u and I_h as:

$$
\frac{I_u}{I_h}=\frac{(\nu_u^a)^2[(\nu_b^a)^2\hat{\alpha}_u-(\nu_u^a)^2\hat{\alpha}_b]^2}{(\nu_h^a)^2[(\nu_b^a)^2\hat{\alpha}_h-(\nu_h^a)^2\hat{\alpha}_b]^2}
\tag{22}
$$

Under the adopted Gamma-family approximation, the allocation rule favors candidates with smaller prior mean objectives or larger prior–simulation discrepancies. It asymptotically satisfies the first-order KKT conditions and provides a stationary ICBA allocation rule. □

Under the adopted Gamma-based approximation, the approximate probability of correct selection (APCS) is given by:

$$
APCS=1-\sum_{u=1,u\neq b}^{N_p}\Phi\left(\frac{\alpha_u\beta_b-\alpha_b\beta_u}{\beta_b\beta_u(\alpha_b+\alpha_u)}\right)
\tag{23}
$$

where $\Phi(\cdot)$ denotes the complementary cumulative distribution function of the standard normal distribution. This derivation is based on the adopted Gamma-family approximation and large shape parameters. Hence, the resulting ICBA rule provides an asymptotic analytical allocation model, rather than an exact characterization of the true scheduling-objective distribution.

REFERENCES

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[4] C.-H. Chen, J. Lin, E. Yücesan, and S. E. Chick, "Simulation budget allocation for further enhancing the efficiency of ordinal optimization," *Discrete Event Dynamic Systems*, vol. 10, no. 3, pp. 251–270, 2000.

SUPPLEMENTARY DATA

TABLE 1
Comparisons among E-HEFT, ILS, RAILS, GA, LGWO, SACS, and CGDR on small-scale benchmark scenarios

Parameter			E-HEFT	ILS		RAILS		GA		LGWO		SACS		CGDR		Wilcoxon test: CGDR vs. GA	
Instance	(n, m)	θ	E^T	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	p-value	Sig(p <0.05)
SFMJ-01	(9,4)	0.01	372.48	236.20	10	225.67	10	202.90	10	199.75	10	**194.57**	10	200.93	10	0.9219	N
SFMJ-01	(9,4)	0.02	370.12	237.35	10	225.87	10	201.19	10	202.64	10	206.98	10	**200.28**	10	0.2754	N
SFMJ-01	(9,4)	0.03	376.20	230.67	10	227.03	10	204.49	10	205.47	10	208.08	10	**199.66**	10	0.0371	Y
SFMJ-01	(9,4)	0.05	358.07	236.92	10	220.50	10	202.77	10	202.92	10	**199.43**	10	202.40	10	0.8457	N
SFMJ-02	(9,4)	0.01	170.57	156.48	10	161.00	10	138.96	10	138.32	10	174.32	10	**136.36**	10	0.2324	N
SFMJ-02	(9,4)	0.02	168.12	156.54	10	163.71	10	138.76	10	**138.41**	10	177.62	10	138.90	10	0.7695	N
SFMJ-02	(9,4)	0.03	171.51	160.46	10	158.89	10	139.67	10	136.14	10	179.51	10	**133.25**	10	0.0371	Y
SFMJ-02	(9,4)	0.05	173.74	158.27	10	160.60	10	141.83	10	135.18	10	174.12	10	**129.11**	10	0.0195	Y
SFMJ-03	(15,4)	0.01	240.60	239.47	10	217.54	10	204.11	10	201.07	10	210.79	10	**199.67**	10	0.0098	Y
SFMJ-03	(15,4)	0.02	240.70	232.32	10	218.97	10	201.43	10	204.76	10	215.14	10	**198.62**	10	0.0273	Y
SFMJ-03	(15,4)	0.03	239.23	233.21	10	225.17	10	212.13	10	203.53	10	207.31	10	**200.77**	10	0.1309	N
SFMJ-03	(15,4)	0.05	239.42	240.93	10	228.60	10	215.59	10	205.26	10	209.71	10	**201.04**	10	0.0488	Y
SFMJ-04	(27,4)	0.01	1316.69	788.33	30	648.22	30	558.67	30	512.89	30	573.45	30	**426.95**	30	0.0098	Y
SFMJ-04	(27,4)	0.02	1371.51	783.53	30	646.82	30	534.87	30	503.82	30	583.48	30	**430.12**	30	0.0020	Y
SFMJ-04	(27,4)	0.03	1349.42	817.16	30	691.64	30	517.15	30	486.12	30	576.20	30	**436.42**	30	0.0195	Y
SFMJ-04	(27,4)	0.05	1353.36	865.00	30	643.84	30	511.16	30	483.70	30	596.57	30	**437.05**	30	0.0137	Y
SFMJ-05	(27,4)	0.01	625.53	543.73	30	540.20	30	344.07	30	355.41	30	354.75	30	**329.17**	30	0.1934	N
SFMJ-05	(27,4)	0.02	626.47	545.99	30	533.24	30	355.55	30	353.79	30	359.23	30	**320.46**	30	0.0273	Y
SFMJ-05	(27,4)	0.03	644.23	535.95	30	526.87	30	337.81	30	336.96	30	358.04	30	**319.30**	30	0.3223	N
SFMJ-05	(27,4)	0.05	639.50	544.67	30	531.17	30	338.69	30	336.09	30	366.03	30	**323.63**	30	0.0195	Y
SFMJ-06	(27,8)	0.01	1030.59	608.86	30	609.81	30	515.08	30	488.35	30	**447.02**	30	451.39	30	0.1602	N
SFMJ-06	(27,8)	0.02	1033.18	656.03	30	569.06	30	497.22	30	501.74	30	460.21	30	**454.59**	30	0.0039	Y
SFMJ-06	(27,8)	0.03	1028.95	675.92	30	548.24	30	523.32	30	491.92	30	465.00	30	**456.65**	30	0.0020	Y
SFMJ-06	(27,8)	0.05	1004.48	657.66	30	599.54	30	512.83	30	504.36	30	**462.46**	30	467.12	30	0.0371	Y
SFMJ-07	(27,8)	0.01	746.87	588.92	30	549.03	30	393.35	30	373.82	30	360.04	30	**353.49**	30	0.1934	N
SFMJ-07	(27,8)	0.02	744.80	545.20	30	589.77	30	395.71	30	377.85	30	368.29	30	**352.45**	30	0.0137	Y
SFMJ-07	(27,8)	0.03	735.56	593.70	30	565.03	30	398.24	30	375.78	30	354.75	30	**344.23**	30	0.0020	Y
SFMJ-07	(27,8)	0.05	742.38	552.36	30	579.16	30	395.69	30	383.72	30	356.82	30	**354.21**	30	0.0039	Y
Average		-	646.94	457.92	-	421.61	-	333.20	-	322.97	-	335.71	-	**299.94**	-	-	-

TABLE 2
Comparisons among E-HEFT, ILS, RAILS, GA, LGWO, SACS, and CGDR on medium-scale benchmark scenarios

Parameter			E-HEFT	ILS		RAILS		GA		LGWO		SACS		CGDR		Wilcoxon test: CGDR vs. GA	
Instance	(n, m)	θ	E^T	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	p-value	Sig(p <0.05)
MFMJ-01	(39,8)	0.01	778.52	699.30	30	583.25	30	437.06	30	398.21	30	425.08	30	**366.24**	30	0.0371	Y
MFMJ-01	(39,8)	0.02	758.98	681.96	30	597.74	30	439.70	30	391.11	30	436.40	30	**355.00**	30	0.0020	Y
MFMJ-01	(39,8)	0.03	755.46	640.14	30	586.42	30	426.97	30	382.73	30	420.84	30	**369.00**	30	0.1055	N
MFMJ-01	(39,8)	0.05	758.29	634.58	30	592.94	30	434.18	30	385.38	30	417.70	30	**363.98**	30	0.0371	Y
MFMJ-02	(39,8)	0.01	792.15	643.93	30	608.41	30	452.40	30	476.37	30	482.28	30	**410.95**	30	0.0098	Y
MFMJ-02	(39,8)	0.02	801.69	631.15	30	598.93	30	458.22	30	489.53	30	476.28	30	**413.46**	30	0.0020	Y
MFMJ-02	(39,8)	0.03	834.42	627.62	30	591.33	30	483.72	30	452.78	30	482.17	30	**411.45**	30	0.0273	Y
MFMJ-02	(39,8)	0.05	781.01	635.93	30	572.12	30	479.97	30	458.49	30	515.24	30	**423.83**	30	0.1309	N
MFMJ-03	(39,8)	0.01	620.34	610.30	30	604.74	30	455.99	30	432.19	30	511.18	30	**413.69**	30	0.3223	N
MFMJ-03	(39,8)	0.02	623.04	593.65	30	582.14	30	470.74	30	451.70	30	508.63	30	**415.14**	30	0.0645	N
MFMJ-03	(39,8)	0.03	634.01	595.23	30	576.15	30	497.05	30	446.79	30	498.05	30	**419.31**	30	0.0039	Y
MFMJ-03	(39,8)	0.05	643.27	572.69	30	583.57	30	488.72	30	435.67	30	499.17	30	**409.79**	30	0.0488	Y
MFMJ-04	(90,16)	0.01	1646.83	1626.81	150	1618.21	150	1508.75	150	1434.67	150	1882.19	150	**1256.88**	150	0.0039	Y
MFMJ-04	(90,16)	0.02	1701.96	1624.18	150	1605.71	150	1546.86	150	1511.28	150	1912.80	150	**1227.40**	150	0.0371	Y
MFMJ-04	(90,16)	0.03	1628.47	1691.37	150	1619.67	150	1582.71	150	1488.53	150	1820.18	150	**1254.89**	150	0.0020	Y
MFMJ-04	(90,16)	0.05	1701.57	1689.61	150	1640.03	150	1509.49	150	1461.01	150	1759.46	150	**1310.88**	150	0.0059	Y
MFMJ-05	(90,16)	0.01	1555.17	1632.57	150	1592.10	150	1467.48	150	1331.43	150	1516.13	150	**1259.78**	150	0.1602	N
MFMJ-05	(90,16)	0.02	1570.23	1638.63	150	1545.93	150	1461.91	150	1361.91	150	1576.49	150	**1263.62**	150	0.3750	N
MFMJ-05	(90,16)	0.03	1608.12	1681.04	150	1592.23	150	1442.42	150	1396.18	150	1475.27	150	**1232.01**	150	0.0098	Y
MFMJ-05	(90,16)	0.05	1566.25	1586.81	150	1558.37	150	1511.88	150	1412.08	150	1533.77	150	**1225.24**	150	0.0020	Y
MFMJ-06	(95,16)	0.01	1185.02	1541.68	200	1488.85	200	1151.75	200	1074.15	200	1391.01	200	**901.50**	200	0.0059	Y
MFMJ-06	(95,16)	0.02	1135.65	1539.66	200	1483.17	200	1121.91	200	1017.60	200	1363.64	200	**873.42**	200	0.0371	Y
MFMJ-06	(95,16)	0.03	1185.18	1553.28	200	1450.65	200	1178.47	200	1037.29	200	1393.51	200	**886.26**	200	0.0020	Y
MFMJ-06	(95,16)	0.05	1143.77	1494.77	200	1463.29	200	1192.64	200	1052.14	200	1413.17	200	**912.00**	200	0.0020	Y
Average	-	-	1100.39	1119.45	-	1072.33	-	925.01	-	865.80	-	1029.61	-	**765.66**	-	-	-

TABLE 3
Comparisons among E-HEFT, ILS, RAILS, GA, LGWO, SACS, and CGDR on large-scale benchmark scenarios

Parameter			E-HEFT	ILS		RAILS		GA		LGWO		SACS		CGDR		Wilcoxon test: CGDR vs. GA	
Instance	(n, m)	θ	E^T	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	E^T	CPU(s)	p-value	Sig(p <0.05)
LFMJ-01	(170,16)	0.01	3905.45	3862.38	500	3510.15	500	3124.05	500	2868.01	500	3137.92	500	**2324.42**	500	0.0020	Y
LFMJ-01	(170,16)	0.02	3716.02	3788.55	500	3548.07	500	3198.79	500	2822.46	500	3233.45	500	**2316.80**	500	0.0020	Y
LFMJ-01	(170,16)	0.03	3962.88	3977.85	500	3640.09	500	2971.86	500	2850.68	500	3198.36	500	**2385.09**	500	0.0020	Y
LFMJ-01	(170,16)	0.05	4024.94	4074.75	500	3928.95	500	2994.51	500	2869.53	500	3310.47	500	**2353.35**	500	0.0020	Y
LFMJ-02	(223,16)	0.01	4327.26	4754.40	500	4385.84	500	3499.33	500	3213.48	500	4083.69	500	**2662.63**	500	0.0020	Y
LFMJ-02	(223,16)	0.02	4460.28	4993.20	500	4563.54	500	3605.00	500	3293.35	500	4170.74	500	**2675.62**	500	0.0020	Y
LFMJ-02	(223,16)	0.03	4360.16	4542.36	500	4553.24	500	3543.42	500	3398.42	500	4253.47	500	**2639.69**	500	0.0020	Y
LFMJ-02	(223,16)	0.05	4202.07	5078.89	500	4849.82	500	3404.77	500	3302.43	500	4300.68	500	**2557.55**	500	0.0020	Y
LFMJ-03	(223,16)	0.01	6835.74	5649.99	500	5077.72	500	3982.20	500	3838.96	500	4574.65	500	**2914.32**	500	0.0020	Y
LFMJ-03	(223,16)	0.02	7233.46	5213.32	500	5221.02	500	3813.60	500	3657.77	500	4407.95	500	**2806.90**	500	0.0020	Y
LFMJ-03	(223,16)	0.03	7582.61	5531.96	500	4965.57	500	3768.50	500	3514.89	500	4391.09	500	**2987.47**	500	0.0020	Y
LFMJ-03	(223,16)	0.05	7013.62	5398.81	500	5102.68	500	3908.03	500	3764.82	500	4479.02	500	**2813.51**	500	0.0020	Y
LFMJ-04	(223,16)	0.01	4429.15	4474.17	500	4249.02	500	3685.57	500	3464.68	500	4512.57	500	**2443.53**	500	0.0020	Y
LFMJ-04	(223,16)	0.02	4821.75	4410.68	500	4281.27	500	3681.06	500	3379.15	500	4600.06	500	**2599.42**	500	0.0020	Y
LFMJ-04	(223,16)	0.03	4553.07	4470.41	500	4251.33	500	3532.14	500	3424.11	500	4442.19	500	**2482.17**	500	0.0020	Y
LFMJ-04	(223,16)	0.05	4697.33	4406.84	500	4164.56	500	3765.77	500	3546.02	500	4592.62	500	**2541.84**	500	0.0020	Y
Average	-	-	5007.86	4664.29	-	4387.05	-	3529.91	-	3325.55	-	4105.56	-	**2594.02**	-	-	-